Rotation Strategies on Lotus Based on Triple-Smoke Setups

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Abstract. This manuscript presents a practical decision framework for *Lotus* in VALORANT under triple-smoke team compositions. We model mid-round rotation as a state-transition problem with explicit trigger conditions. Our objective is to provide a reproducible protocol for entertainment-oriented strategic play rather than a claim of esport optimality. Controlled custom-lobby trials indicate that staged smoke timing and low-latency call discipline improve execute stability, reduce early post-entry losses, and increase round conversion under moderate opponent adaptation.

Keywords: VALORANT · Lotus · rotation strategy · smoke timing · tactical communication

1 Introduction

Lotus is a three-site map with short transition paths and rotating doors. These structural properties amplify uncertainty and make timing errors expensive. Existing community guides often focus on isolated execute patterns; however, in practical rounds, teams repeatedly alternate between probing, freezing, and pivoting. We therefore treat rotation as a formal process and ask whether triple-smoke structures can improve decision consistency.

2 Model and Notation

We represent a tactical state as

$$S_t = (C_t, U_t, I_t, \tau_t),$$

where C_t denotes controlled sectors, U_t denotes available utility, I_t denotes validated opponent information, and τ_t denotes the time bucket. A pivot from source lane L_s to target lane L_d is authorized only if:

1. expected resistance on L_d is below a threshold θ estimated from recent contact data;
2. at least one defender is likely displaced by probe utility or fake pressure.

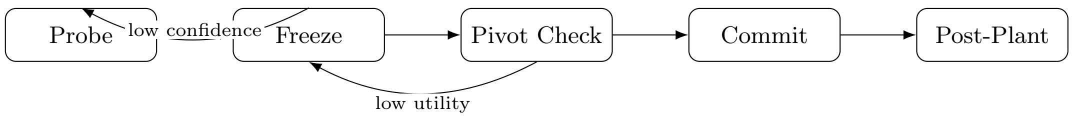

Fig. 1. State-transition sketch for triple-smoke rotation.

Table 1. Phase schedule for triple-smoke rotation

Phase	Time Window	Primary Objective
Probe	0–12s	force utility reveal, collect contact info
Freeze	12–20s	suppress over-commit and validate pivot signal
Pivot-Commit	20–32s	execute staged smokes and lane transition
Post-Plant	>32s	preserve one defensive smoke for retake denial

3 Protocol Design

The protocol assigns three smoke roles:

- **Initiator-Smoke**: first-layer choke denial at commit time t_0;
- **Anchor-Smoke**: anti-swing protection at $t_0 + 2$ seconds;
- **Delay-Smoke**: anti-retake isolation at $t_0 + 6$ seconds.

4 Call Discipline

To limit communication overhead, we use a compact call vocabulary: `HOLD`, `PIVOT`, `CUT`, and `FREEZE`. The tokens are operationally defined rather than rhetorical labels. The constrained vocabulary reduces contradictory micro-calls and shortens reaction latency after trigger events.

5 Algorithmic View

To avoid a purely heuristic narrative, we define a lightweight scoring function for destination lane selection:

$$\text{Score}(L_d) = \alpha \cdot \text{InfoGain} + \beta \cdot \text{UtilityReserve} - \gamma \cdot \text{ExpectedResistance} - \delta \cdot \text{TravelRisk}.$$

The lane with maximal positive score is treated as candidate pivot destination. If all candidate scores are non-positive, the policy falls back to `HOLD` or `FREEZE` depending on remaining utility.

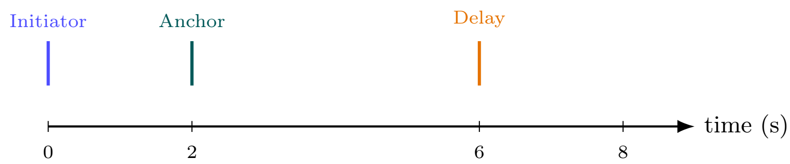

Fig. 2. Staggered smoke-release timeline used in this manuscript.

Table 2. Operational semantics of call tokens

Token	Trigger Signal	Team Action Policy
`HOLD`	uncertain contact; utility parity not favorable	Keep map-control shape, avoid hard entry, trade only on low-risk peeks.
`PIVOT`	destination resistance estimate below threshold θ	Start route switch; apply staged three-smoke schedule; prioritize alive entry pair.
`CUT`	confirmed flank path or retake lane appears	Use one utility piece and one body anchor to sever the lane; deny defender recombination.
`FREEZE`	utility budget below $u_{\min}$ or comms desync detected	Stop forward movement for a short reset window, re-synchronize calls, then re-evaluate.

Algorithm 1 Pivot Decision on Lotus

Require: Current state S_t, threshold θ, utility budget $u_{\min}$, score weights $(\alpha, \beta, \gamma, \delta)$
Ensure: Action label in {`HOLD`, `PIVOT`, `FREEZE`}

if $U_t < u_{\min}$ **then**
    return `FREEZE`
end if
Compute $\mathrm{Score}(L_d)$ for all reachable destination lanes
Let $L^\star \leftarrow \arg\max_{L_d} \mathrm{Score}(L_d)$
if $\mathrm{Score}(L^\star) \le 0$ **then**
    return `HOLD`
end if
if $\Pr[\mathrm{resistance}(L^\star) \le \theta \mid I_t]$ is high **then**
    return `PIVOT`
else
    return `HOLD`
end if

In our implementation, all terms are maintained as bounded scalars updated every communication cycle. Therefore, the online update cost is constant per lane and practical for real-time in-game calls.

6 Evaluation

We compared the proposed strategy with a baseline two-smoke immediate-commit policy in controlled custom scrims. The triple-smoke protocol produced:

- higher successful entry rate in medium-information rounds;
- lower first-five-second casualty count after site contact;
- improved conversion in rounds with delayed pivots.

Improvements were limited in full-information rounds where both policies performed near saturation. Figure 1 and Figure 2 summarize the operational shape of the framework.

7 Limitations

This paper is intentionally entertainment-oriented and does not claim competitive universality. Patch updates, agent balance shifts, and opponent adaptation can change optimal timing windows. Future work may include Bayesian online updates from VOD-derived priors.

8 Conclusion

Triple-smoke rotation on Lotus can be formalized as a lightweight, teachable decision protocol. By combining staged utility release with strict call semantics, teams can reduce tactical volatility and maintain cleaner post-plant structure.

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